# Life Platform

## Personal Health Intelligence System

*19 health apps. All excellent. None of them talking to each other. So I built the layer that joins them.*

| 30 | 122 | 19 | 9 | ~$3/mo | v2.80.1 |
|---|---|---|---|---|---|
| LAMBDAS | MCP TOOLS | DATA SOURCES | EMAILS / WEEK | AWS COST | VERSION |

M. Walker · 2026 · Built across ~80 AI-assisted development sessions

*INTERNAL USE — Sensitive data obfuscated throughout*

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# The Story

Why this exists — and why it matters beyond personal health

## The Problem

I like data. Over time I had accumulated 19 health and wellness apps — Whoop, Eight Sleep, CGM, Garmin, Withings, MacroFactor, Notion, Habitify, and more. Each one excellent at its job. None of them talking to each other.

The problem wasn't lack of data — it was lack of synthesis. **What does all of this actually mean, together?** Each app reported its number in isolation. The real insights — the ones that cross sources, apply personal baselines, and produce actionable answers — lived in the gaps between them. No tool on the market filled those gaps. So I built one.

## Why Not Buy a Solution?

No existing platform crosses all 19 sources, applies personal baselines, runs custom algorithms (PhenoAge from my actual blood draws, TSB from my actual training load), and delivers results as AI-coached narrative. The closest consumer tools integrate 3–4 sources at best. The closest enterprise tools cost $50k+/year and aren't designed for individuals.

The only path was to build it. And building it from scratch meant every architectural decision, every AI pattern, every ops failure was *mine to own*.

DUAL PURPOSE

### Personal lab.

Real stakes. Real data. Real consequences. The platform isn't a prototype — it runs every morning against live health data.

### Enterprise stress-test.

Every AI feature — personas, extraction, adaptive emails, error handling — mirrors a pattern relevant to enterprise Claude adoption. The learning transfers directly.

| 19 sources | 4 years | ~80 sessions | $3/month | 0 days | v2.80.1 |
|---|---|---|---|---|---|
| unified into one | of health data | with Claude to build | fully operational | manual data entry | current version |

PART ONE

# The Solution

What it does · how it fits into daily life · what it tells you

This section is about the experience — what the platform actually does for you, day-to-day, and what kinds of analysis it makes possible that no single app can provide.

## 01 — A Week in the Life

Monday to Sunday — how everything connects into daily progress

The platform doesn't require anything different from your day. Every device you're already wearing syncs automatically. The only thing that changes is what's waiting in your inbox when you wake up.

| MON | TUE | WED | THU | FRI | SAT | SUN |
|---|---|---|---|---|---|---|
| **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily | **Morning Brief** 10 AM daily |
| **Day Grade** A–F + context | **Day Grade** A–F + context | **Day Grade** A–F + context | **Day Grade** A–F + context | **Day Grade** A–F + context | **Day Grade** A–F + context | **Day Grade** A–F + context |
| **Anomaly?** conditional alert | | **Chronicle** Elena publishes | **Levelled up?** XP + reward | **The Plate** food col + list | | **Weekly Digest** 7-pillar review |
| | | | | | | **Brittany Email** partner update |
| **API Sync** background | **API Sync** background | **API Sync** background | **API Sync** background | **API Sync** background | **API Sync** background | **API Sync** background |
| **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM | **Dashboard** 2 PM + 6 PM |

**Every morning**

The Daily Brief lands before the first coffee. Sleep architecture, yesterday's scores, training detail, nutrition gaps, habit streaks, readiness signal, and an AI coaching note from the Board of Directors — all from data that collected itself overnight.

**If something's off**

The anomaly detector runs at 8:05 AM across 15 metrics and 7 sources. If two or more metrics diverge from your personal baseline on the same morning, you get an alert with a root-cause hypothesis — before the daily brief even fires.

**Wednesday**

Elena Voss — an AI persona — publishes 'The Measured Life' as a weekly health narrative. It draws on journal entries, key metrics, and coaching moments to tell a story about what actually happened. It exists because numbers don't create behaviour change — narrative does. The blog also has a voice section: an audio version where Elena actually speaks the column, so you can listen to your week rather than just read it.

**Friday**

The Weekly Plate: a food column reviewing the week's meals, recipe riffs on what worked, and a personalised grocery list for the local supermarket built from the actual MacroFactor nutrition log.

**Sunday**

Two accountability outputs land. First: the Weekly Digest — a full 7-pillar health review. Second: two accountability signals. The Brittany email is a warm, narrative update written by Dr. Murthy (relationships) and Coach Rodriguez (behaviour) — it answers 'how is he actually doing?' for a partner who cares but doesn't track metrics. Separately, an accountability dashboard sends signals to close friends who are on their own health journeys — green/yellow/red indicators based on key metrics, so if texting goes quiet they know whether to check in or whether things are on track and no nudge is needed.

## 02 — Why Apps Aren't Enough

The problem with siloed data — and what unified analysis actually gives you

Every app you use gives you one answer. Whoop tells you your recovery score. MacroFactor tells you your calories. Eight Sleep tells you your sleep score. Notion holds your journal. They don't talk to each other — so none of them can answer the only question that actually matters:

***"My HRV is down, my sleep score is off, and my recovery is low. Why — and what should I actually do about it?"***

Answering that question properly requires pulling from Whoop (HRV baseline), Eight Sleep (bed temperature, restlessness), MacroFactor (alcohol, caloric deficit, meal timing), Garmin (training load from yesterday), Notion (journal entry — were you stressed?), Habitify (sleep hygiene habits), and Apple Health (late-night phone use proxy). That's seven sources. No app does that join. This platform does.

| Question | What Whoop says | What this platform can say |
|---|---|---|
| Why was my sleep poor? | *Sleep score: 54* | Bed temp was +2°F, you ate within 90 min of sleep, alcohol logged in MacroFactor, and your journal noted stress |
| Should I train hard today? | *Recovery: 42% (yellow)* | TSB is -18 (fresh), HRV within 5% of baseline, no anomalies flagged, yesterday was rest — train hard |
| Why is my glucose elevated? | *No glucose data* | CGM: high-carb meal last night (MacroFactor), sleep was fragmented (Eight Sleep), possible cortisol signal |
| Is my training improving my fitness? | *VO2 max estimate* | Zone 2 HRV trend over 8 weeks, pace-at-HR regression, lactate threshold proxy from Strava + Garmin |
| What's affecting my mood most? | *No mood data* | Journal enrichment × HRV × sleep × training load — which variable has the highest Pearson r with mood score |

## 03 — What No Single App Can Tell You

Cross-source analysis, personal baselines, and derived insights that require joining all 19 sources

Each domain below represents a cluster of analysis that goes beyond what any single app provides — because it crosses sources, applies personal baselines, or derives a metric that no app computes for you.

**WHY CUSTOM CALCULATIONS MATTER**

Whoop's recovery score uses population averages. MacroFactor's calorie target uses estimated TDEE. Garmin's fitness age is a model, not your data. Every metric below is computed against your personal baseline, your actual lab draws, your real glucose response curves — not generalised population stats.

### Sleep — the full picture

- **Sleep efficiency vs bed environment:** does your Eight Sleep set-point actually correlate with better sleep for you, specifically?
- **Alcohol → sleep architecture:** your personal dose-response curve across 100+ nights — not population averages
- **Fasting glucose validation:** CGM overnight nadir vs. lab draws — are your morning blood tests actually fasting values?
- **Circadian consistency:** sleep onset standard deviation — are your 1 AM nights actually costing you next-day HRV?

### Training — what's actually working

- **Zone 2 aerobic base trend:** pace-at-same-HR over time. Improving fitness = same pace, lower HR.
- **Lactate threshold proxy:** cardiac efficiency curve from steady-state Garmin + Strava sessions — closest consumer proxy to a lab test
- **Heart rate recovery trend:** 1-minute and 2-minute HR drop post-workout — strongest exercise-derived mortality predictor (Cole et al., NEJM)
- **Training load vs. readiness:** TSB from CTL/ATL, Whoop recovery, Garmin Body Battery — synthetic recommendation rather than three separate scores
- **Exercise variety:** Shannon diversity index across movement patterns — staleness detection, missing category identification

### Nutrition — beyond macro tracking

- **Meal-level glucose response:** CGM × MacroFactor food log — which specific foods cause spikes for you (not population data)
- **Energy balance:** Apple Watch TDEE (measured) vs. MacroFactor intake — actual daily surplus/deficit, not estimated
- **Protein distribution score:** how many meals hit the MPS threshold (30g)? Timing matters as much as total
- **Micronutrient sufficiency:** fiber, potassium, magnesium, vitamin D, omega-3 — surfaced daily, not just macros
- **Genome × nutrition:** FADS2 ALA conversion, VKORC1 vitamin K sensitivity, choline demand — your specific gene panel informs food priorities

### Mental & emotional health

- **Mood × physiology:** journal mood scores correlated with HRV, sleep, training load — which physical variable predicts your mood best?
- **Defense mechanism detection:** 11 patterns (intellectualization, avoidance, displacement) identified in journal text by Claude, trending over time
- **Social connection quality:** meaningful-vs-surface interaction ratio, isolation risk detection, Murthy threshold assessment (3–5 close relationships)
- **Emotional depth index:** journal vocabulary richness, flow state indicators, values-lived alignment — week-on-week

### Longevity & metabolic health

- **Biological age (PhenoAge):** Levine algorithm from 9 blood biomarkers across all 7 lab draws — with genome context for longevity SNPs
- **Metabolic health score:** composite (0–100) from CGM 30%, labs 35%, weight/BMI 20%, blood pressure 15% — one number, not four dashboards
- **ASCVD 10-year risk:** Pooled Cohort Equations from actual blood draws — tracked across 6 years (2019–2025)
- **Health trajectory:** forward projection across weight, biomarkers, Zone 2 fitness, recovery, metabolic trends

## Gamification — keeping the loop closed

- **Character Level (1–100):** 7 weighted pillars composited into a persistent level score — not a number to minimise, but one to genuinely improve
- **Cross-pillar effects:** Sleep Drag (-3 to Metabolism when sleep <6hr), Training Boost (+2 when Zone 2 >150 min/wk), Synergy Bonus
- **Pixel art avatar:** body frame morphs with physical milestone events; tier aura evolves from Foundation to Elite
- **Rewards system:** user-defined rewards triggered by milestone events — seeded by Matthew and Brittany together
- See next page for screenshots of the avatar, radar chart, and tier progression system

## 04 — Why 124 Tools

The architecture of inquiry — one question, many sources

The number looks large. It makes sense when you understand the structure.

Every meaningful health question crosses multiple data sources. Each source needs its own read tool. Each cross-source analysis needs its own correlation or synthesis tool. Each derived metric (biological age, metabolic score, lactate proxy) needs its own compute tool. Add write tools for logging (supplements, travel, life events) and operation tools (experiments, rewards, board management) — and 124 is actually compact for what it covers.

**THE KEY ARCHITECTURAL CONSTRAINT**

Claude has no memory between sessions. Every conversation starts from zero.

This is why there are 122 tools — not because the data is complex, but because each question requires discrete retrievals across multiple sources. Fewer tools = less precise answers.

| Category | Tools | Examples |
|---|---|---|
| Source read tools | 38 | get_sleep_analysis · get_training_load · get_cgm_dashboard · get_mood_trend |
| Cross-source correlations | 22 | alcohol → sleep · exercise → glucose · mood × HRV · meal → spike |
| Derived analytics | 18 | biological_age · metabolic_score · lactate_threshold · hr_recovery |
| Gamification / tracking | 12 | character_sheet · pillar_detail · experiment_results · habit_tier_report |
| Write / logging tools | 16 | log_supplement · log_travel · log_ruck · log_temptation · save_insight |
| Platform / config management | 18 | board_of_directors · update_member · adaptive_mode · get_profile |

The other reason: Claude has no memory between sessions. Every tool call is a discrete retrieval. If Claude needs to answer "how does your sleep change when you train in the evening?" it issues four tool calls: get_sleep_analysis, get_exercise_sleep_correlation, get_training_load, get_day_type_analysis. Fewer tools would mean less precise answers — or no answer at all.

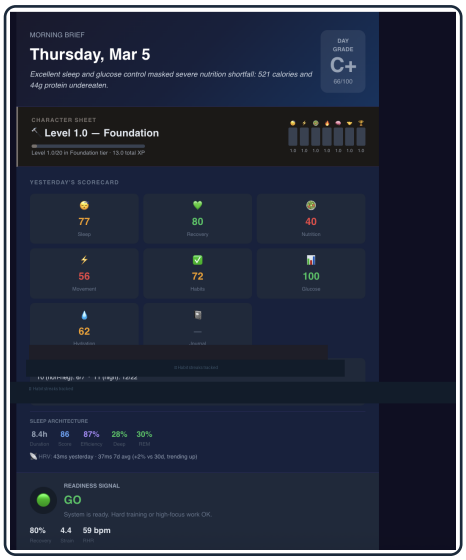
| Module | Tools | Domain |
|---|---|---|
| tools_sleep | 8 | Sleep quality, environment, onset consistency, nap tracking |
| tools_activity | 10 | Zone 2, training load, HR recovery, lactate proxy, periodization |
| tools_nutrition | 9 | Food log, macro targets, meal glucose, energy balance, micronutrients |
| tools_character | 4 | Character sheet, pillar detail, level history, effects |
| tools_habits | 8 | Habit registry, tier report, vice streaks, synergy groups |
| tools_journal | 6 | Entries, search, mood trend, correlations, insights, defense patterns |
| tools_social | 12 | Life events, interaction log, temptation, cold/heat exposure |
| tools_longevity | 8 | Biological age, metabolic score, health risk, genome, food response DB |

| | | |
|---|---|---|
| tools_correlation | 8 | Alcohol × sleep, exercise × glucose, mood × vitals, jet lag |
| tools_health | 10 | Readiness, blood pressure, gait, body comp, trajectory, day type |
| tools_labs | 8 | Lab results, trends, out-of-range history, DEXA, biomarker search |
| tools_board | 3 | Board of directors CRUD — add/update/remove expert personas |
| (14 more modules) | 30 | cgm · genome · supplements · weather · travel · adaptive mode · ops … |

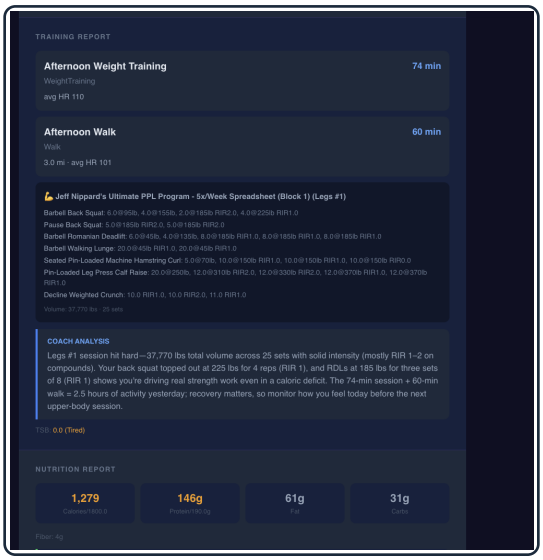
# 05 — Features in Action

Screenshots from the running system — obfuscated for sharing

## Daily Brief — every morning at 10 AM

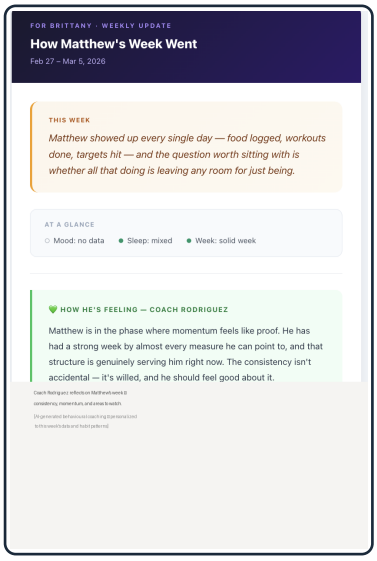



*Top of the Daily Brief: day grade, Character Sheet level + radar, 8-pillar scorecard*

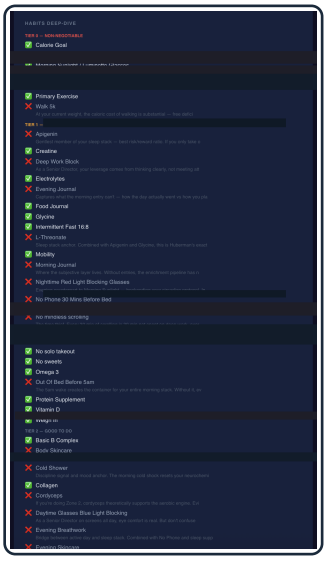



*Training report section: set-level workout data + AI sports scientist commentary*

## Brittany Email (Sunday 9:30 AM) · Habits Deep-Dive




*Partner email: Rodriguez on behavioural state, Dr. Murthy on how to show up*




*Habits deep-dive: 65 habits across 3 tiers with AI coaching note per habit*

# The Weekly Plate · Buddy Page · CGM + Board

Food column · accountability beacon · data intelligence

## The Weekly Plate

Friday Edition · March 2, 2026

### This Week on Your Plate

What a week, Matthew. You've been riding a serious wave — down 4.4 pounds in the last seven days, sitting at 292.9 now. The last few days have been a masterclass in protein efficiency: those Morning Smoothies (which you've clearly committed to), the return of your epic seed-and-nut yogurt bowls, and some seriously creative dinners. Thai basil ground meat over rice? Mongolian beef? Japanese curry katsu? You're not just hitting macros — you're actually cooking food you want to eat. The early-week grilled chicken and quinoa rhythm gave way to more adventurous flavors, and it's working. Let's build on that momentum.

### Your Greatest Hits

**Morning Smoothies (6 days running)**

Your new non-negotiable. 80 calories, 15g protein, and it's clearly anchoring your mornings. This is the kind of habit that compounds. Keep a backup canister at the office — you're never missing this again.

**The Seed & Nut Yogurt Bowl**

Nonfat Greek yogurt + chia, flax, hemp, walnuts, pecans, berries. You logged this 4 times. It's ~650 cal, 35g protein, and genuinely satisfying. The only tweak? You could swap half the nuts for more berries on lighter days — same volume, fewer calories. But honestly, if it ain't broke.

**Lean Ground Beef (93%) + Quinoa + Spinach**

Your weeknight closer. Five times in two weeks. ~700 cal, 60g+ protein, done in 25 minutes. This is the kind of reliable dinner that lets you experiment elsewhere. Pro move: cook the quinoa in low-sodium chicken broth next time — same macros, way more flavor.

**Olive Oil (your cooking fat MVP)**

Logged 9 times. You're using it right — a tablespoon here, a drizzle there. Just remember it's 120 cal per tablespoon, so measure when you can. Met Market has a great Spanish olive oil in the specialty section that's worth the splurge.

*The Weekly Plate: reviewing the week's meals + recipe riffs on what actually worked*

## The Grocery Run

**Protein**

- Lean ground beef, 93% (your staple — grab 2 lbs)
- Chicken breast, boneless skinless (for the katsu bowls)
- Flank steak (for chimichurri bowls)
- Cottage cheese, 2% (Good Culture or Nancy's — the wildcard)
- Eggs, large (running low?)

**Produce**

- Fresh parsley (for chimichurri)
- Cucumber (for Turkish bowls)
- Lemon (for Turkish bowls + general use)
- Garlic (for chimichurri)
- Spinach, fresh (your go-to green)
- Red bell peppers (you've been using these)
- Jalapeños (for your egg scrambles)
- Onions, yellow (your cooking base)
- Blueberries (for yogurt bowls)
- Blackberries (for yogurt bowls)
- Coleslaw mix, bagged (for katsu bowls)

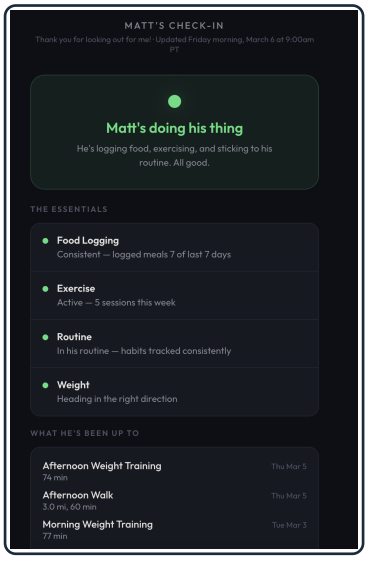
**Dairy & Eggs**

- Greek yogurt, nonfat plain (your bowl base — 2 large tubs)
- Cottage cheese, 2% (see above!)
- Eggs (if needed)

**Pantry & Staples**

- Quinoa, dry (you're going through this quickly)
- White rice (for katsu bowls)
- Panko breadcrumbs (you already have some, but check)
- Olive oil (always)
- Sesame oil (for katsu slaw)
- Rice vinegar (for katsu slaw)
- Red wine vinegar (for chimichurri)
- Cumin, ground (for Turkish bowls)
- Paprika, smoked (for Turkish bowls)
- Cinnamon, ground (for Turkish bowls)
- Red pepper flakes (for chimichurri)
- Chia seeds (running low?)
- Flax seeds (running low?)
- Hemp seeds (running low?)
- Walnuts (running low?)
- Pecans (running low?)
- Vanilla protein powder (check your stash)
- Everything bagel seasoning (for cottage cheese)

*The Grocery Run: grocery list generated from the MacroFactor nutrition log*



MATT'S CHECK-IN

Thank you for looking out for me! · Updated Friday morning, March 6 at 9:00am PT

**Matt's doing his thing**

He's logging food, exercising, and sticking to his routine. All good.

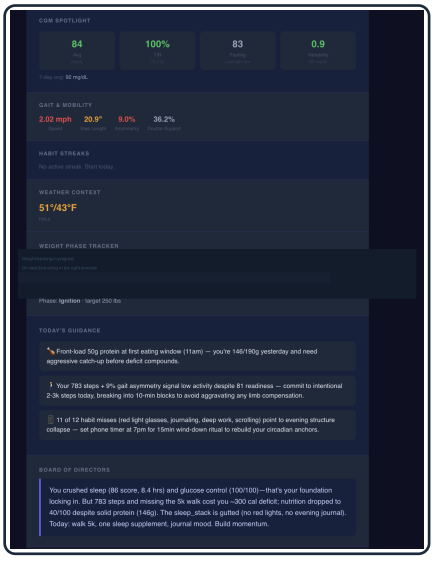
THE ESSENTIALS

- **Food Logging** — Consistent — logged meals 7 of last 7 days
- **Exercise** — Active — 5 sessions this week
- **Routine** — In his routine — habits tracked consistently
- **Weight** — Heading in the right direction

WHAT HE'S BEEN UP TO

| | |
|---|---|
| Afternoon Weight Training<br>74 min | Thu Mar 5 |
| Afternoon Walk<br>3.0 mi, 60 min | Thu Mar 5 |
| Morning Weight Training<br>77 min | Tue Mar 3 |

*Buddy page: green/yellow/red accountability beacon for Tom (Singapore) to check*



CGM SPOTLIGHT

| Avg | TIR | Variability | Mendelian |
|---|---|---|---|
| 84 | 100% | 83 | 0.9 |

7-day avg: 92 mg/dL

GAIT & MOBILITY

| Speed | Step Length | Asymmetry | Double Support |
|---|---|---|---|
| 2.02 mph | 20.9" | 9.0% | 36.2% |

HABIT STREAKS

No active streak. Start today.

WEATHER CONTEXT

51°/43°F

WEIGHT PHASE TRACKER

Phase: **Ignition** · target 250 lbs

TODAY'S GUIDANCE

- Front-load 50g protein at first eating window (11am) — you're 146/180g yesterday and need aggressive catch-up before deficit compounds.
- Your 783 steps + 9% gait asymmetry signal low activity despite B1 readiness — commit to intentional 2-3k steps today, breaking into 10-min blocks to avoid aggravating any limb compensation.
- 11 of 12 habit misses (red light glasses, journaling, deep work, scrolling) point to evening structure collapse — set phone timer at 7pm for 15min wind-down ritual to rebuild your circadian anchors.

BOARD OF DIRECTORS

You crushed sleep (86 score, 8.4 hrs) and glucose control (100/100)—that's your foundation locking in. But 783 steps and missing the 5k walk cost you ~300 cal deficit; nutrition dropped to 40/100 despite solid protein (146g). The sleep_stack is gutted (no red lights, no evening journal). Today: walk 5k, one sleep supplement, journal mood. Build momentum.

*CGM spotlight + gait + weight phase + Board of Directors coaching card*

# Dashboard · Blog · Clinical Summary

Live web properties — all generated automatically

*Web dashboard: readiness, sleep, glucose, training TSB, character avatar — updates 2×/day*

THE MEASURED LIFE

A weekly chronicle by Elena Voss

Archive About

*"Before the Numbers"*

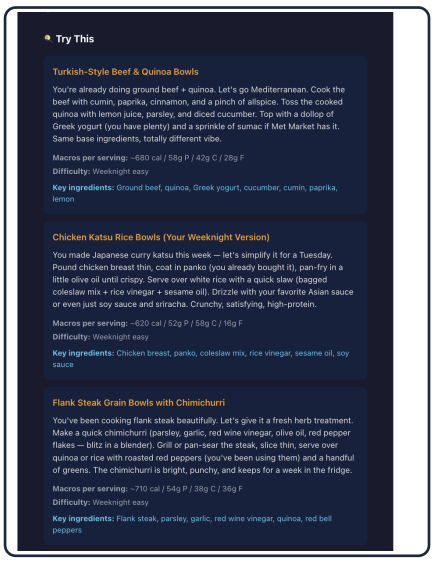
Prologue · February 2026 · Seattle, WA

The first thing you notice about Matthew's apartment is the charging cables.

There's one snaking from the nightstand — that's the Whoop strap, a fitness tracker that monitors his heart rate variability while he sleeps. Another on the bathroom counter feeds a Garmin Epix watch that measures his stress levels every three seconds. The Eight Sleep pod cover on his bed regulates temperature through the night and grades his sleep architecture each morning. His phone syncs a continuous glucose monitor adhered to the back of his arm. A Withings scale in the bathroom measures not just weight but body composition, fat mass, lean mass, the whole topography of a body in transition.

Nineteen data sources in total. Over a hundred metrics tracked daily. All feeding into a system he built himself — not an app you can download, but a custom health intelligence platform running on Amazon Web Services, ingesting data from every sensor on and around his body, processing it through algorithms he designed during weekends and late nights, and delivering a morning brief to his inbox before he's finished his first coffee.

*'The Measured Life' — weekly health narrative by AI persona Elena Voss, auto-published*



**Try This**

**Turkish-Style Beef & Quinoa Bowls**

You're already doing ground beef + quinoa. Let's go Mediterranean. Cook the beef with cumin, paprika, cinnamon, and a pinch of allspice. Toss the cooked quinoa with lemon juice, parsley, and diced cucumber. Top with a dollop of Greek yogurt (you have plenty) and a sprinkle of sumac if Met Market has it. Same base ingredients, totally different vibe.

**Macros per serving:** ~680 cal / 58g P / 42g C / 28g F

**Difficulty:** Weeknight easy

**Key ingredients:** Ground beef, quinoa, Greek yogurt, cucumber, cumin, paprika, lemon

**Chicken Katsu Rice Bowls (Your Weeknight Version)**

You made Japanese curry katsu this week — let's simplify it for a Tuesday. Pound chicken breast thin, coat in panko (you already bought it), pan-fry in a little olive oil until crispy. Serve over white rice with a quick slaw (bagged coleslaw mix + rice vinegar + sesame oil). Drizzle with your favorite Asian sauce or even just soy sauce and sriracha. Crunchy, satisfying, high-protein.

**Macros per serving:** ~620 cal / 52g P / 58g C / 16g F

**Difficulty:** Weeknight easy

**Key ingredients:** Chicken breast, panko, coleslaw mix, rice vinegar, sesame oil, soy sauce

**Flank Steak Grain Bowls with Chimichurri**

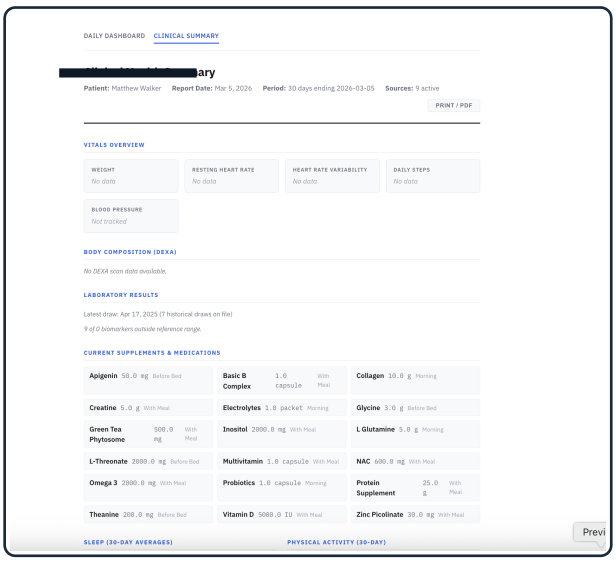
You've been cooking flank steak beautifully. Let's give it a fresh herb treatment. Make a quick chimichurri (parsley, garlic, red wine vinegar, olive oil, red pepper flakes — blitz in a blender). Grill or pan-sear the steak, slice thin, serve over quinoa or rice with roasted red peppers (you've been using them) and a handful of greens. The chimichurri is bright, punchy, and keeps for a week in the fridge.

**Macros per serving:** ~710 cal / 54g P / 38g C / 36g F

**Difficulty:** Weeknight easy

**Key ingredients:** Flank steak, parsley, garlic, red wine vinegar, quinoa, red bell peppers

*Recipe suggestions tuned to personal macro history and local supermarket availability*



DAILY DASHBOARD CLINICAL SUMMARY

[redacted]ary

**Patient:** Matthew Walker **Report Date:** Mar 5, 2026 **Period:** 30 days ending 2026-03-05 **Sources:** 9 active

PRINT / PDF

VITALS OVERVIEW

| WEIGHT | RESTING HEART RATE | HEART RATE VARIABILITY | DAILY STEPS |
|---|---|---|---|
| *No data* | *No data* | *No data* | *No data* |

| BLOOD PRESSURE |
|---|
| *Not tracked* |

BODY COMPOSITION (DEXA)

*No DEXA scan data available.*

LABORATORY RESULTS

Latest draw: Apr 17, 2025 (7 historical draws on file)

*9 of 0 biomarkers outside reference range.*

CURRENT SUPPLEMENTS & MEDICATIONS

| | | |
|---|---|---|
| **Apigenin** 50.0 mg Before Bed | **Basic B Complex** 1.0 capsule With Meal | **Collagen** 10.0 g Morning |
| **Creatine** 5.0 g With Meal | **Electrolytes** 1.0 packet Morning | **Glycine** 3.0 g Before Bed |
| **Green Tea Phytosome** 500.0 mg With Meal | **Inositol** 2000.0 mg With Meal | **L Glutamine** 5.0 g Morning |
| **L-Threonate** 2000.0 mg Before Bed | **Multivitamin** 1.0 capsule With Meal | **NAC** 600.0 mg With Meal |
| **Omega 3** 2000.0 mg With Meal | **Probiotics** 1.0 capsule Morning | **Protein Supplement** 25.0 g With Meal |
| **Theanine** 200.0 mg Before Bed | **Vitamin D** 5000.0 IU With Meal | **Zinc Picolinate** 30.0 mg With Meal |

SLEEP (30-DAY AVERAGES) PHYSICAL ACTIVITY (30-DAY)

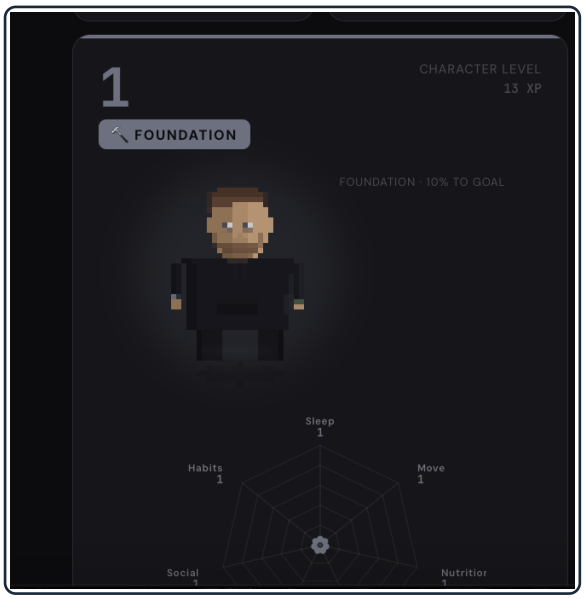
Previ

*Clinical Summary: 30-day report formatted for a doctor visit (name field redacted)*

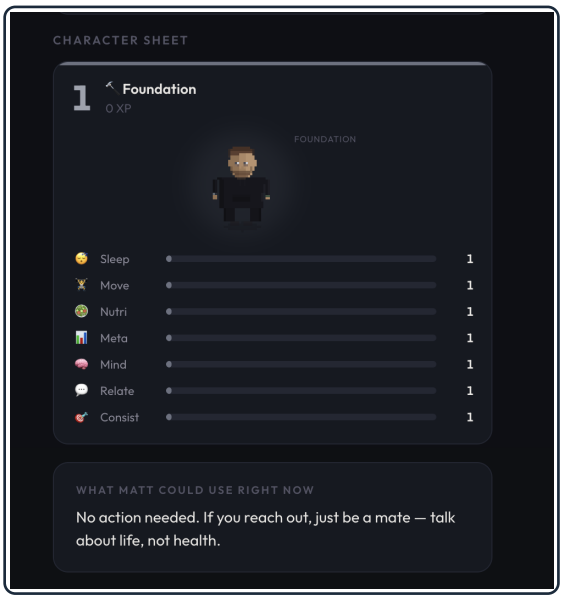
# Gamification — Avatar · Radar · Tier Progression

The Character Sheet lives on the web dashboard, the buddy page, and in every daily brief

## How it looks in the running system




*Web dashboard: Level 1 Foundation avatar + 7-pillar radar chart (Sleep, Move, Habits, Social, Nutrition, Mind, Meta)*




*Buddy page character sheet: pixel art avatar + 7 pillar bars — what Tom sees in Singapore*

## Five Tiers — Foundation to Elite

Level progresses through five tiers. Each tier changes the avatar's aura, unlocks new pillar effects, and shifts the coaching tone. The XP bar fills based on weighted pillar improvement using a 21-day EMA. Tier transitions take 7 days up and 10 days down — preventing gaming and rewarding consistency.

| Mechanic | Detail |
|---|---|
| Level range | 1–100 · baseline Feb 22 2026 · EMA λ=0.85 over 21 days |

| | |
|---|---|
| Tier bands | Foundation (1–20) · Momentum (21–40) · Discipline (41–60) · Mastery (61–80) · Elite (81–100) |
| Transition speed | 5 consecutive days above threshold to move up · 7 to move down |
| XP calculation | Weighted pillar scores: Sleep 20% · Movement 18% · Nutrition 18% · Mind 15% · Metabolic 12% · Consistency 10% · Relationships 7% |
| Cross-pillar effects | Sleep Drag (-3 pts when sleep <6hr) · Training Boost (+2 pts when Zone 2 >150 min/wk) · Nutrition Synergy · Habit Streak Bonus |
| Avatar body frames | 4 milestone frames composited via CSS — reflects physical progress milestones |
| Pillar badge overlays | 7 badge PNGs overlay on the avatar sprite — one per pillar, glow on high scores |
| Pixel art tier aura | 5 aura variants (none → grey → gold → red → teal → purple) — rendered on dashboard + buddy page |
| Reward examples (Ph.4) | Weekend away · Spa day · New gear (fitness equipment, clothing) · Dinner out · Concert tickets · Experience-based events · Seeded together by Matthew + Brittany |

PART TWO

# The Build

Architecture · AWS · security · data model · resiliency

This section covers how it's built — the AWS infrastructure, deployment patterns, data model, and operational practices that keep it running reliably for ~$3/month.

# 06 — System Architecture

Ingest → Store → Serve · matplotlib-rendered, not guessed

Five layers: Data Sources (19 sources) → Ingest (EventBridge + 30 Lambdas + API Gateway + SQS DLQs) → Store (DynamoDB + S3 + Secrets Manager + CloudWatch + CloudTrail) → Serve/Intelligence (MCP Lambda + 7 Email Lambdas + Anthropic API + Dashboard Refresh + Ops) → Outputs (SES · 3 CloudFront sites · Claude Desktop/Mobile · AWS Budget).

# 07 — AWS Components

What's running and why

| Service | Usage | Count | Notes |
|---|---|---|---|
| Lambda | Ingest + compute + email + MCP + ops | 30 | Python 3.12 · 128–1024 MB |
| DynamoDB | Single-table health data store | 1 table | On-demand · PITR · deletion protection |
| S3 | Raw archive + static site + config JSON | 1 bucket | ~2.5 GB · 4 path prefixes |
| EventBridge | Cron scheduler for all Lambdas | 22 rules | UTC crons; DST shift documented in RUNBOOK |
| API Gateway | Inbound webhook from iPhone (HAE) | 1 HTTP | Route POST /ingest → Lambda |
| CloudFront | CDN for 3 web properties | 3 distros | Lambda@Edge auth on dash + buddy |
| SES | Outbound email + inbound insight capture | 1 domain | 9 email types/week |
| Secrets Manager | API keys + OAuth tokens | 6 secrets | 90-day auto-rotation on MCP key |
| CloudWatch | Monitoring + alerting | 35 alarms | → SNS → email |
| SQS | Dead-letter queues on async Lambdas | 20+ DLQs | Async failure capture |
| ACM | TLS certificates | 2 certs | us-east-1 required for CloudFront |
| AWS Budget | Cost guardrail | $20/mo | Alerts at 25% / 50% / 100% · current: ~$3/mo |

```
You a[redacted]on has entered the ALARM state,
(05/0[redacted]

View this alarm in the AWS Management Console:
https://us-west-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-west-2#alarmsV2:alarm/life-platform-ingestion-failures

Alarm Details:
- Name:                       life-platform-ingestion-failures
- Description:
- State Change:               OK -> ALARM
- Reason for State Change:    Threshold Crossed: 1 datapoint [1.0 (05/03/26 21:42:00)] was greater than or equal to the threshold (1.0).
- Timestamp:                  [redacted] UTC
- AWS Account:                [redacted]
- Alarm Arn:                  [redacted]failures

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.
```

*CloudWatch alarm email (account ID redacted) — real production alert, resolved < 1hr*

```
⚠️ Life Platform: Stale Data Detected

The following sources have not updated in over 48 hours:

  - Todoist tasks: Last update: 2026-03-04 (65h ago)

Full source status:
  ✅ Whoop recovery/sleep: 2026-03-06 (17h ago)
  ✅ Withings weight/body comp: 2026-03-05 (41h ago)
  ✅ Strava activities: 2026-03-05 (41h ago)
  ⚠️  Todoist tasks: 2026-03-04 (65h ago)
  ✅ Apple Health: 2026-03-05 (41h ago)
  ✅ Eight Sleep: 2026-03-05 (41h ago)
  ✅ MacroFactor nutrition: 2026-03-05 (41h ago)
  ✅ Garmin biometrics: 2026-03-05 (41h ago)
  ✅ Habitify habits: 2026-03-05 (41h ago)

Checked at: 2026-03-06 16:45 UTC
```

*Freshness checker: daily staleness report — catches data gaps before the morning brief*

# 08 — Security & Ops

IAM · secrets · auth · git deployment · cost guardrails

## IAM — Least-Privilege Roles

| Role | Permissions |
|---|---|
| Ingestion (×13) | DynamoDB PutItem · S3 PutObject · SecretsMgr (own secret) · SQS DLQ |
| MCP server | DynamoDB GetItem/Query/PutItem (cache) · S3 GetObject (cgm/* only) |
| Email / digest | DynamoDB read/write · SES SendEmail (own domain) · S3 dashboard/* |
| CloudFront auth | SecretsMgr GetSecretValue · CF viewer request only |

## Auth Layers

| Surface | Method |
|---|---|
| dash.[domain] | Lambda@Edge HMAC cookie |
| buddy.[domain] | Separate Lambda@Edge + secret |
| blog.[domain] | Public / no auth |
| MCP endpoint | Bearer token (HMAC-SHA256) |
| HAE webhook | Bearer + Secrets Manager |
| Insight email | ALLOWED_SENDERS whitelist |

## Secrets Management

- 6 secrets (consolidated from 12 — saves $2.40/mo)
- OAuth tokens: self-healing refresh on each Lambda invocation
- MCP API key: 90-day auto-rotation via dedicated rotator Lambda

## Cost Controls

- $20/month hard cap · 3-tier alerts (25% / 50% / 100%)
- On-demand DynamoDB — no reserved capacity
- Lambda memory right-sized: 128–256 MB for ingestion
- **Current: ~$3/month** for full workload

---

## Git-Driven Deployment

The entire platform lives in a single git repository (v2.80.1.1 — 741 objects, ~80 sessions of commits). Every Lambda, every config file, every deploy script is versioned.

### Deploy Pattern

- **1. Write** — Claude edits files via Filesystem tools
- **2. Commit** — git commit -m 'v2.X.X: feature description'
- **3. Deploy** — deploy/deploy_lambda.sh auto-reads handler config from AWS
- **4. Verify** — CloudWatch logs checked before sign-off
- **5. Push** — git push; GitHub is source of truth

### Why This Matters

- deploy_lambda.sh never hardcodes zip filenames — handler config read live from AWS to avoid mismatch
- 10s wait between sequential Lambda deploys prevents ResourceConflictException
- Deploy success ≠ execution success — CloudWatch log verification is required
- Session close ritual: write handover → update CHANGELOG → git add -A && commit && push

---

## Data Governance & Privacy

All health data remains in a single AWS account (us-west-2) under exclusive control. No third-party analytics, no data sharing, no advertising. Data flows: device APIs → Lambda (ephemeral, no persistent storage) → DynamoDB/S3 (encrypted at rest). Sensitive values (API keys, OAuth tokens) never appear in Lambda environment variables — all retrieved from Secrets Manager at invocation time and held only in memory. The obfuscation pipeline (demo mode) removes or replaces all PII and health values before any data leaves the system for sharing.

# 09 — Data Model

Single-table DynamoDB · 21 source partitions · one clean pattern

## Key Pattern

**PK — Partition Key**

```
USER#[id]#SOURCE#<source>
```

**SK — Sort Key**

```
DATE#YYYY-MM-DD
```

No GSI by design. All access patterns served by PK+SK range queries. Monthly aggregation for date windows >90 days.

## Special Partitions

| Pattern | Purpose |
|---|---|
| PROFILE#v1 | User settings, habit registry, demo rules |
| CACHE#[id] / TOOL# | Pre-computed tool results (TTL 26h) |
| SOURCE#day_grade | Daily composite scores — 948+ records |
| SOURCE#character_sheet | RPG level + 7 pillars (computed nightly) |
| SOURCE#insights / INSIGHT# | Coaching insights captured via email reply |

## 21 Source Partitions

| Source | Domain | Type |
|---|---|---|
| whoop | sleep / HRV | Wearable API |
| eightsleep | bed environment | Smart bed API |
| garmin | biometrics | Wearable API |
| strava | cardio | Activity API |
| withings | body comp | Scale API |
| macrofactor | nutrition | CSV/Dropbox |
| apple_health | steps/gait/cgm | Webhook 4h |
| habitify | habits | API |
| notion | journal | Notion API |
| labs | biochemical | Manual seed |
| dexa | body comp | Manual seed |
| genome | genetic | Manual seed |
| supplements | supplements | MCP write |
| weather | environment | Open-Meteo |
| state_of_mind | emotional | Webhook |
| character_sheet | gamification | Computed |
| day_grade | meta score | Computed |
| habit_scores | habits meta | Computed |
| anomalies | monitoring | Computed |
| travel | context | MCP write |

**SOURCE-OF-TRUTH DOMAIN MODEL**

When the same signal arrives from multiple sources, a single source is declared authoritative per domain. **Whoop** = sleep metrics and HRV (wrist sensor, location-independent). **Eight Sleep** = bed environment only (temperature, mattress score). **Garmin** = active biometrics during workouts. **Strava** = canonical activity record (synced from Garmin). **MacroFactor** = nutrition (TDEE-adjusted). This prevents conflicting scores and eliminates double-counting in aggregated metrics.

# 10 — AI Integration

Claude MCP · 122 tools · OAuth 2.1 · Sonnet 4.6 for synthesis · Haiku for extraction

## MCP Architecture

- Lambda Function URL + OAuth 2.1 auto-approve flow
- HMAC-SHA256 Bearer token validation in handler
- Streamable HTTP transport (MCP spec 2025-06-18)
- Claude Desktop · claude.ai · Claude mobile — all connected
- Cold start: ~700–800ms / Warm: ~25ms
- 12 tools pre-computed nightly (cache hit <100ms)
- 5-min in-memory cache for Board of Directors config

## AI Model Standards

| Task | Model |
|---|---|
| Synthesis / narrative | Claude Sonnet 4.6 |
| Daily coaching email | Claude Sonnet 4.6 |
| Extraction / classify | Claude Haiku 4.5 |
| Anomaly hypothesis | Claude Haiku 4.5 |
| Cost per brief | ~$0.02 |
| Plate email / week | ~$0.04 |

## AI Calls Per Daily Brief

| Call | Model | Output |
|---|---|---|
| Board of Directors | Sonnet | Expert persona coaching from health data |
| Training + Nutrition | Sonnet | Commentary from sports scientist + nutritionist |
| Journal Coach | Sonnet | Reflection + action from raw journal text |
| TL;DR + Guidance | Haiku | 2-sentence summary + top 3 recommendations |

## Board of Directors — The Concept

**WHY PERSONAS, NOT PROMPTS?**

A recovery score of 42 means different things depending on who you ask. To a sleep scientist it's a HRV signal. To a behavioural coach it's a stress response. To a nutritionist it's a refuelling window.

Personas force the AI to frame the same data through different lenses — each with different expertise, vocabulary, and priorities. The result is richer, more actionable coaching than any single prompt can produce.

**Config-driven: edit any persona in S3 without redeploying a Lambda.**

13 expert AI personas: Dr. Lisa Park (sleep), Dr. Sarah Chen (training), Dr. Marcus Webb (nutrition), Dr. Peter Attia (metabolic), Coach Rodriguez (behavioral), Dr. Paul Conti (psychiatry), Dr. Vivek Murthy (social), Elena Voss (narrator).

# 11 — Resiliency

DLQs · alarms · gap-aware backfill · 3 incidents resolved <1hr each

## Dead-Letter Queues

20 of 30 Lambdas have SQS DLQs on async invocations. Failed runs are captured — not silently lost.

## Gap-Aware Self-Healing Backfill

- On each run: query DDB for last 7 days of records
- Identify any missing DATE# keys for each source
- Fetch only those specific days from upstream API
- Normal run: 1 DDB query, 0 extra API calls
- Applied to all 6 API-based ingestion Lambdas

## Hard-Won Operational Notes

- **Deploy success ≠ execution success.** CloudWatch log verification required
- **S3 AccessDenied can mask NoSuchKey.** Without ListBucket, root cause is obscured
- **DynamoDB 'date' is reserved.** Requires expression attribute name substitution
- **ACM certs for CloudFront must be in us-east-1** regardless of bucket region
- **MCP tool functions must come before TOOLS={} dict** — order matters at import

## CloudWatch Alarms (35)

| Type | Count |
|---|---|
| Lambda Error rate | 30 (1 per Lambda) |
| SQS DLQ depth > 0 | 3 |
| Lambda duration > 80% | 2 |

All alarms → SNS → email. 24hr window, TreatMissingData=notBreaching.

## 3 Production Incidents — All Resolved <1hr

- **Secrets consolidation:** stale SECRET_NAME env var on 2 Lambdas — CloudWatch surfaced immediately, env var patched
- **CloudFront double-pathing:** origin path '/dashboard' caused 403 on avatar assets — URL path analysis resolved in <20 min
- **Handler filename mismatch:** Weekly Digest used wrong zip entry — CloudWatch import error on first invocation, corrected in one redeploy

# 12 — Documentation System

Changelog · handovers · incident log · RCA · runbooks — all version-controlled

An AI-dependent system that restarts from zero each session needs a different kind of documentation. Every doc in this repo serves a specific function — not bureaucracy, but continuity infrastructure.

## Document Registry

| Document | Purpose |
|---|---|
| HANDOVER_LATEST.md | Current state snapshot: version, open threads, last decisions. Read at session start. |
| CHANGELOG.md | Every version bump with what changed, what was fixed, and why. 80+ entries. |
| INCIDENT_LOG.md | All production incidents with severity, TTD, TTR, data loss status, and root cause. |
| docs/rca/ | Full Root Cause Analysis per incident. Structured: timeline, contributing factors, fix, prevention. |
| RUNBOOK.md | Step-by-step ops: deploy a Lambda, rotate secrets, trigger backfill, reset alarms. |
| ARCHITECTURE.md | System design, data flow diagrams, source-of-truth domain model. |
| SCHEMA.md | DynamoDB key patterns, field names, reserved word workarounds. |
| MCP_TOOL_CATALOG.md | All 124 MCP tools: description, inputs, source data, last updated. |
| FEATURES.md | Every shipped feature with status, version, and design notes. |

## The Handover System

~80 Claude sessions, ~80 context resets. The handover file is the continuity layer — written at the end of every session, read at the start of the next. Without it, every session restarts from zero. With it, each session builds on the last.

| | |
|---|---|
| **125+** handover files | **106+** CHANGELOG entries |
| **24+** incidents logged | **3** full RCAs written |

## Incident Discipline

Every production incident is logged with severity (P1–P4), time to detect, time to resolve, and whether data was lost. P1/P2 incidents get a full RCA document with contributing factors, timeline, fix applied, and prevention added.

| Level | Definition |
|---|---|
| P1 | System broken — no data flowing, MCP completely down |
| P2 | Major feature broken, data loss risk, or multi-day gap |
| P3 | Single source affected — degraded but functional |
| P4 | Cosmetic, minor data quality, or transient error |

***Enterprise lens:*** *AI pipelines fail silently — output looks plausible even when broken. Incident logging + structured RCA is the discipline that catches it.*

# 13 — Roadmap

What's next

## Near-Term

| Feature | Status | What it adds |
|---|---|---|
| Monarch Money | Auth done | Financial data as health input: spend, savings rate, net worth alongside health scores |
| Reward System (Ph. 4) | Partial | User-defined rewards triggered by Character Sheet milestones — weekend away, spa day, new gear, experience events |
| Conversational Coach | Designed | iOS Shortcut → voice check-in → Lambda → Claude → structured journal entry |
| Google Calendar | Designed | Demand-side context: meeting load, travel days, and schedule density as health inputs |
| LLM Failover Router | Planned | If Anthropic API is unavailable, a router Lambda retries with OpenAI GPT-4o as fallback — keeps all 9 weekly emails firing even during outages |

## Intelligence Layer

| Feature | Description |
|---|---|
| Predictive Anomaly | Prophet/scikit-learn personal forecasting: 'HRV is 15% lower than expected given training load' |
| Causal Inference Engine | Bayesian causal graph — does exercise cause better sleep, or vice versa? (DoWhy library) |
| Digital Twin / Simulation | 'What if I increased Zone 2 to 4hr/week?' Projects outcomes from personal response curves |
| Semantic Search | Embeddings across journal, Chronicle, Board commentary — find every mention of a topic |

## Open-Source Framework

- Config-driven Lambda ingest templates (any OAuth2 API → same pattern)
- Single-table DynamoDB health schema with source-of-truth domain model
- MCP tool scaffolding + Board of Directors persona pattern
- Character Sheet engine (any weighted pillar set, any domain)
- The Chronicle blog engine (any AI narrator persona)

## 14 — Key Learnings

Patterns from ~80 sessions of building with AI

### One person can go end-to-end

Design, infrastructure, code, testing, deployment, documentation — all with Claude as the engineering partner. For greenfield builds, AI-assisted development doesn't just speed things up — it changes what's feasible for one person.

*Enterprise lens: Small-team AI pilots are more viable than you think. The bottleneck isn't headcount — it's context and iteration speed.*

### Handover docs are infrastructure

~80 sessions means ~80 context resets. HANDOVER_LATEST.md is the most important file in the repo. Without it, every session restarts from zero. Writing it consistently compounds — by v2.80 it contains a precise snapshot of every open thread.

*Enterprise lens: AI-assisted teams need session continuity protocols. The equivalent of a handover doc is a standard your org should define before scaling.*

### Configuration over code

Board of Directors personas, Character Sheet weights, source-of-truth domains — all S3 JSON config. No Lambda redeploy to change an expert voice or adjust a pillar weight. The right default for anything policy-like.

*Enterprise lens: AI persona and prompt governance should be config, not code — so non-engineers can tune model behaviour without touching a deployment pipeline.*

### Observability from day one

35 alarms, 30 DLQs, structured logging, and a freshness checker — all built alongside features, not retrofitted. All 3 production incidents resolved <1hr because the signal was already there.

*Enterprise lens: AI pipelines need the same instrumentation discipline as any service. Silent AI failures are worse than silent code failures — the output looks plausible.*

### $3/month is a design constraint, not a side effect

Every resource choice evaluated against a $20/month hard cap. Serverless + on-demand is the right default for this invocation class. The constraint made the architecture better.

*Enterprise lens: Cost ceilings force good serverless design. A $20/mo personal project and a $200k/yr enterprise workload share the same patterns.*

### The dual-use value was unplanned

Every AI feature — personas, adaptive emails, gamification — also tested something enterprise-relevant: prompt engineering under real constraints, structured extraction at scale, graceful error handling under uncertainty. The overlap wasn't designed in. It's just what building real things with AI looks like.

*Enterprise lens: The best AI training for your team is building something real — not a workshop, not a sandbox. Real stakes surface the real problems.*

---

**The system is live right now.**

Everything in this presentation came from a live system. Ask anything — I can query it via Claude right now.

M. Walker · 2026